TOR AMPLITUDE OVER THE POLYNOMIAL PRESENTATION RING LIFTS FROM THE FIBRE FOR PSEUDO-COHERENT COMPLEXES

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Allowed standard prerequisites (editorial).

- A pseudo-coherent complex over A = R[x_1, ..., x_d]/I with A flat over R is pseudo-coherent as a complex of R[x_1, ..., x_d]-modules. invoked at proof lines 22193-22195. Reduces to the polynomial presentation ring.
- For a map R_p -> A_q of local rings with A_q a quotient of a regular local ring, a pseudo-coherent complex whose fibre has cohomology confined to [a, b] is perfect with tor amplitude in [a - d, b]. invoked at proof lines 22196-22199. Produces the perfectness and the shifted tor amplitude over A_q.
- A flat ring map pushes tor amplitude bounds through: a perfect complex over A_q with tor amplitude in [a - d, b] over A_q has tor amplitude in [a - d, b] over R_p. invoked at proof lines 22200-22201. Transfers the tor amplitude down to R.

Source reference. Original lemma 0E1T: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21885–21885. Cited at proof lines 22193-22194 to reduce to the polynomial presentation; the conclusion is used inside the counted span.

Source reference. Original lemma 09PC: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20229–20229. Cited at proof lines 22196-22199 to produce perfectness with tor amplitude in [a - d, b] over A_q; the conclusion is used inside the counted span.

Source reference. Original lemma 066J: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17100–17100. Cited at proof lines 22200-22201 to transfer the tor amplitude to R; the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let $A = R[x_1, \ldots, x_d]/I$ be flat and of finite presentation over R. Let $\mathfrak{q} \subset A$ be a prime ideal lying over $\mathfrak{p} \subset R$. Let $K \in D(A)$ be pseudo-coherent. Let $a, b \in \mathbf{Z}$. If $H^i(K_\mathfrak{q} \otimes^{\mathbf{L}}_{R_\mathfrak{p}} \kappa(\mathfrak{p}))$ is nonzero only for $i \in [a, b]$, then $K_\mathfrak{q}$ has tor amplitude in $[a - d, b]$ over R.*

Proof. By Lemma 0E1T K is pseudo-coherent as a complex of $R[x_1, \ldots, x_d]$-modules. Therefore we may assume $A = R[x_1, \ldots, x_d]$. Applying Lemma 09PC to $R_\mathfrak{p} \to A_\mathfrak{q}$ and the complex $K_\mathfrak{q}$ using our assumption, we find that $K_\mathfrak{q}$ is perfect in $D(A_\mathfrak{q})$ with tor amplitude in $[a - d, b]$. Since $R_\mathfrak{p} \to A_\mathfrak{q}$ is flat, we conclude by Lemma 066J. □

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